Amazon.com, Inc. (AMZN) Quantitative Case Study

HISTORICAL TIME-SERIES ANALYSIS & FINANCIAL PROBLEM SET (1997 – 2025)

Dataset Overview: Daily open, high, low, close, adjusted close, and volume data for Amazon.com, Inc. spanning from May 15, 1997, through September 14, 2025. Total observations exceed 7,000 trading days across multiple macroeconomic cycles.

1. Dataset Schema & Descriptive Statistics

Examine the structure of the dataset variables provided in `AMZN_1997-05-15_2025-09-14.csv` before attempting the problem set below.

Column Name	Data Type	Description
Date	YYYY-MM-DD	Trading session date from IPO era to Q3 2025.
Open / High / Low	Float (USD)	Intraday price boundaries unadjusted for splits.
Close / Adj Close	Float (USD)	Closing price and split/dividend-adjusted closing price.
Volume	Integer	Total number of shares traded during the session.

2. Problem 1: Daily Return & Volatility Modeling

Exercise 1.1: Log Returns and Annualized Volatility

Using the `Adj Close` column, calculate the daily logarithmic return $r_t = \ln(P_t / P_{t-1})$. Assuming 252 trading days in a year, compute the annualized historical volatility $\sigma_{ann} = \sigma_{daily} \times \sqrt{252}$ for three distinct epochs:

- **Epoch A:** Dot-Com Era (1997-05-15 to 2002-12-31)
- **Epoch B:** Post-Crisis Expansion (2010-01-01 to 2019-12-31)
- **Epoch C:** Post-Pandemic & Rate Hiking Era (2020-01-01 to 2025-09-14)

Deliverable: Provide a comparative table of annualized volatilities and discuss structural shifts in risk profile.

3. Problem 2: Moving Average Crossover Strategy Backtest

Exercise 2.1: Simple Moving Average (SMA) Optimization

Implement a quantitative backtest for a dual moving average crossover strategy. Let fast window k_1 = 50 days and slow window k_2 = 200 days. Generate a buy signal when $SMA_{50} > SMA_{200}$ and a sell signal when $SMA_{50} < SMA_{200}$.

- Calculate total strategy return versus a Buy-and-Hold benchmark over the full 28-year timeline.
- Compute the Sharpe Ratio assuming a risk-free rate r_f = 2.5% annualized.

4. Problem 3: Maximum Drawdown & Crisis Analysis

Exercise 3.1: Identifying Peak-to-Trough Declines

Maximum Drawdown (MDD) measures the maximum observed loss from a peak to a trough of a portfolio, before a new peak is attained. Using the `Adj Close` series:

- Identify the exact start date, trough date, and recovery date for Amazon's two largest historical drawdowns (Dot-Com crash 2000–2001 and the 2022 macroeconomic correction).
- Calculate the percentage drawdown formula: MDD = (Trough Value − Peak Value) / Peak Value × 100%.

5. Submission Guidelines & Grading Rubric

Solutions must be submitted as a reproducible Jupyter Notebook or Python script utilizing `pandas` and `numpy`.

Evaluation Criterion	Weight	Description
Data Cleaning & Alignment	20%	Handling missing dates, corporate actions, and correct indexing.
Algorithmic Correctness	40%	Accurate vectorization of returns, moving averages, and drawdown formulas.
Analytical Commentary	40%	Depth of financial insight regarding risk-adjusted returns and market regimes.